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## Course Delivery Overview

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WEBD 3100 — Web Design Fundamentals is a hands-on, project-based course designed to develop practical skills in web interface design, usability, responsive design, prototyping, implementation, accessibility, and evaluation.

Throughout the course, you will learn how to take website requirements and user needs, organize content, establish a visual direction, develop interface designs in Figma, create interactive prototypes, translate those designs into a working website, and evaluate and refine the completed interface.

The course follows a practical design-to-production process:

**Understand → Plan → Design → Prototype → Implement → Deploy → Evaluate → Refine → Deliver**

Design concepts are introduced progressively and then reinforced through workshops, design activities, critique, project milestones, implementation, usability and accessibility evaluation, and a semester-long Website Redesign Project.

### Design-to-Production Workflow

From the beginning of the course, you will use a professional workflow that connects interface design with web development and deployment.

Three primary technologies support this workflow:

**Figma → Git/GitHub → GitHub Pages**

**Figma** is used to develop and communicate interface designs, including style guides, design systems, wireframes, responsive layouts, reusable components, and interactive prototypes.

**Git and GitHub** are used to manage the implementation of your designs, maintain your source code, and provide a history of development and refinement.

**GitHub Pages** is used to deploy your website so that it can be viewed, tested, evaluated, and shared as a working web experience.

Together, these tools support the workflow:

**Design → Develop → Version → Deploy → Verify**

Week 1 includes **Workshop 0 — Course Tooling & Environment Setup**, where you will configure and verify Figma, Git/GitHub, GitHub Pages, and the development environment required for the course.

# Building Your Web Design Skills

The course begins by examining what makes a website effective.

You will consider the relationship between **purpose, audience, content, usability, accessibility, visual communication, and technical implementation**.

You will then explore visual design principles including hierarchy, balance, alignment, proximity, repetition, contrast, whitespace, colour, and typography.

You will also learn **Nielsen's 10 Usability Heuristics** and begin using them as a framework for analyzing web interfaces. This provides a more structured way to explain why an interface works well, where users may experience difficulty, and how a design could be improved.

As the course progresses, you will use Figma to develop style guides and design systems, organize website content, create sitemaps, develop responsive wireframes, produce high-fidelity interfaces, and build interactive prototypes.

You will then move from design into implementation by translating your approved Figma designs into responsive HTML and CSS.

Later in the semester, you will evaluate the implemented website for accessibility and usability, identify problems, make evidence-based improvements, and complete quality assurance before final production delivery.

By the end of the course, you should be able to move from a website requirement or design problem to a complete and defensible web solution.

## Workshops

Workshops provide regular opportunities to apply course concepts through hands-on, interactive activities.

Workshops are intended to support a **learning journey**, rather than simply provide a collection of small assignments.

Before many workshops, you will participate in presentations, demonstrations, discussions, and analysis activities. These activities establish the underlying theory and help determine what you already understand about the topic before beginning the practical work.

A typical workshop learning cycle may include:

**Prior-Knowledge Check → Presentation / Demonstration → Discussion → Workshop Analysis → Hands-on Activity → Collaboration → Feedback → Reflection**

You are encouraged to discuss problems, compare approaches, critique designs, test ideas, troubleshoot issues, and learn collaboratively with other students.

The instructor will use discussion and workshop activity to identify different levels of understanding within the class and provide additional explanation, examples, demonstrations, or guidance where required.

Each workshop concludes with a **required individual reflection**. Reflection gives you an opportunity to identify what you learned, explain important decisions, consider feedback, recognize challenges, and describe

how the learning can be applied to your semester project.

Workshop activities progress through website critique, visual design and usability heuristics, colour and typography, information architecture, UI prototyping, design-to-code implementation, accessibility remediation, and final design and implementation quality assurance.

## Design Critique and Usability Heuristics

Critique is an important part of learning web design.

At the beginning of the course, you will examine existing websites and discuss what you believe works well and what could be improved.

Rather than relying only on statements such as *"I like this design"* or *"this looks better,"* you will progressively develop a professional vocabulary for explaining design decisions.

You will learn to evaluate interfaces using visual design principles and **Nielsen's 10 Usability Heuristics**.

As your knowledge develops, you should increasingly be able to explain:

**What is the design trying to accomplish?**

**Who is the intended user?**

**What does the user need to notice or do?**

**What design principles are being applied?**

**Where could the user experience difficulty?**

**What evidence supports the recommended change?**

These skills will later be applied to the evaluation of your own implemented website.

## Semester Website Redesign Project

A major component of WEBD 3100 is the **Semester Website Redesign Project**.

Rather than beginning a large website project near the end of the semester, you will progressively design, prototype, implement, test, and refine the project as your skills develop.

The project follows the same design-to-production process used throughout the course:

**Define → Organize → Wireframe → Design → Prototype → Build → Deploy → Evaluate → Refine → Deliver**

The project is divided into four major milestones.

**Milestone 1 — Design Brief + Figma Style Guide** establishes the project requirements and visual direction. You will begin defining the design language of the website through elements such as colour, typography, spacing, imagery, and interface components.

**Milestone 2 — Sitemap + Responsive Figma Wireframes** establishes the structure of the website. You will organize content, develop navigation and page hierarchy, and plan how important pages will work across

mobile and desktop layouts.

**Milestone 3 — Responsive UI Design + Interactive Figma Prototype** transforms the structural design into a high-fidelity interface. You will apply your design system, develop responsive layouts and reusable components, and create an interactive prototype demonstrating important user journeys.

**Milestone 4 — Usability Test, Findings & Recommendations** evaluates the implemented website. You will apply usability heuristics and structured user testing to identify usability problems, document findings, and recommend evidence-based improvements.

Each milestone builds toward the final Website Redesign Project.

## From Figma to Code

An important transition occurs during the second half of the course when the project moves from interface design into production implementation.

Your Figma design becomes a specification for the website you will build.

You will translate your interface designs into semantic HTML and maintainable CSS while applying responsive layout techniques such as CSS Grid and Flexbox.

The objective is not simply to create another website that contains similar content.

The implementation should demonstrate a clear relationship between the approved design and the working interface.

You should be able to compare:

### **Figma Design ↔ Implemented Website**

and explain how design decisions involving typography, colour, spacing, hierarchy, navigation, responsive behaviour, and interface components were translated into code.

Git and GitHub will be used throughout implementation, and the working website will be deployed through GitHub Pages.

## Accessibility

Accessibility is considered part of good web design rather than an activity performed only after development is complete.

You will consider accessibility throughout design and implementation and later perform a more focused accessibility evaluation of the working interface.

You will examine areas such as:

- semantic structure
- keyboard accessibility
- focus states
- colour contrast
- alternative text

- accessible navigation
- forms
- responsive behaviour
- common accessibility barriers

You will use testing and evaluation to identify accessibility problems and then remediate the implementation.

The objective is to develop the habit of considering accessibility as part of normal design and development practice.

## Usability Testing & Evaluation

Later in the semester, you will return to the usability principles introduced earlier in the course.

You will apply **Nielsen's 10 Usability Heuristics** and structured user testing to evaluate your implemented website.

This provides an opportunity to compare your original design intentions with how people actually experience the interface.

You will document observations, identify usability problems, analyze findings, and develop recommendations for improvement.

The process can be summarized as:

**Observe → Identify → Analyze → Recommend → Improve → Verify**

Usability testing therefore becomes part of the development process rather than simply an evaluation performed after the website is finished.

## Refinement and Quality Assurance

A website is not complete simply because it has been implemented.

During the final stages of the course, you will use the findings from accessibility evaluation, usability testing, peer feedback, and instructor feedback to refine your project.

Quality assurance activities include:

- responsive testing
- cross-browser testing
- content review
- link validation
- accessibility remediation
- usability remediation
- design-system consistency
- Figma-to-implementation comparison
- final interface refinement

You will be encouraged to approach problems systematically:

**What was intended?**

**What was implemented?**

**What did testing reveal?**

**What needs to change?**

**Did the change solve the problem?**

This creates an iterative process of testing and improvement before final delivery.

## Final Website Redesign Project

During the final weeks of the course, you will integrate and refine the work completed throughout the semester into a production-ready Website Redesign Project.

Your final submission will demonstrate the complete design-to-production journey through three connected artifacts:

**Figma** — what you designed and prototyped

**GitHub** — how you implemented and managed the project

**GitHub Pages** — the working website experienced by the user

Your final project should demonstrate your ability to apply:

- web and visual design principles
- usability heuristics
- colour and typography
- style guides and design systems
- information architecture
- responsive wireframing
- high-fidelity interface design
- interactive prototyping
- semantic HTML and maintainable CSS
- responsive web implementation
- accessibility
- usability testing
- quality assurance
- Git and GitHub development practices
- GitHub Pages deployment

Week 14 is the **FINAL WEEK REVIEW & Project Completion** period. You will use this time to compare your Figma designs with the implemented website, perform final testing, resolve accessibility and usability issues, improve design fidelity, complete quality assurance, and prepare the project for final delivery.

In Week 15, you will submit your **Final Website Redesign Project** and complete the **Final Delivery & Design Rationale**.

The final delivery provides an opportunity to demonstrate the completed website and explain the decisions and evidence that shaped its development.

# Learning Through Design, Critique and Iteration

Good web design rarely emerges from a single attempt.

Design involves making decisions, testing assumptions, receiving feedback, discovering problems, and improving the solution.

Throughout WEBD 3100, you will be encouraged to treat critique and testing as part of the design process.

When evaluating your work, you should increasingly learn to ask:

**What problem am I trying to solve?**

**Who am I designing for?**

**What evidence supports this design decision?**

**Does the interface communicate what I intended?**

**Can users accomplish what they need to accomplish?**

**What did accessibility and usability testing reveal?**

**What should I change as a result?**

By the end of WEBD 3100, the goal is not simply to create an attractive website. You should be able to take requirements and user needs, develop and justify an interface design, prototype it in Figma, translate it into a responsive implementation, deploy it, evaluate it for accessibility and usability, and progressively refine it into a professional web experience.

The complete learning journey can be summarized as:

**Understand → Plan → Design → Prototype → Implement → Deploy → Evaluate → Refine → Deliver**